

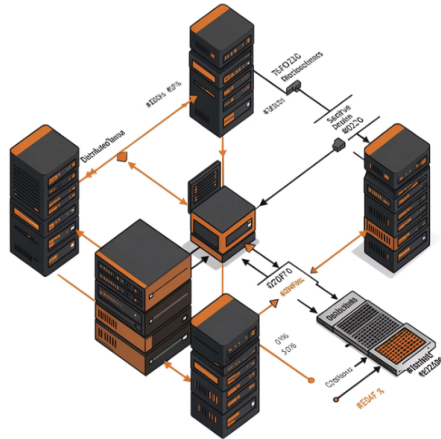


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Cloud Resource Allocation Optimization

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The Challenge



Core Problem

Given **N tasks** with varying processing times and **K servers**, assign each task to exactly one server such that the **maximum load on any server is minimized**.

Constraints

- Tasks must be assigned **sequentially** — order is preserved
- Each task assigned to exactly one server
- No task splitting allowed
- All tasks must be completed

Real-World Interpretation: Think of this as distributing jobs across worker threads, containers, or cloud instances without reordering the queue.

Binary Search on Answer

Rather than searching for an assignment directly, we **binary search the answer space** — the possible values of maximum load — and use a greedy check to validate each candidate.



Define Space

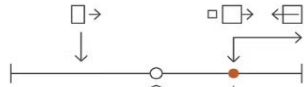
Binary Search

Greedy Check

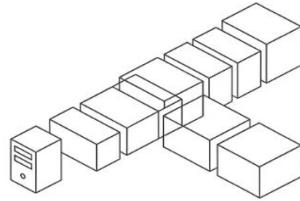
Converge

The search space is bounded: **low** = **max(task)** (at least one server handles the largest task) and **high** = **sum(tasks)** (one server handles everything). The greedy feasibility check runs in $O(N)$, giving overall **$O(N \log N)$** time complexity.

Greedy Feasibility Check



Binary Search Phase
(Narrows answer range,
 $O(\log N)$ iterations, tests
candidate max loads)



Greedy Check Phase
(Assigns tasks left-to-
spills to next server
when capacity exceeded,
 $O(N)$ per check)

How isPossible() Works

For a candidate max load X , iterate through tasks and accumulate load on the current server. When adding the next task would exceed X , spill to the next server.

- Count servers used — must be $\leq K$
- If feasible, try a smaller max load
- If infeasible, increase the candidate

Time

$O(N \log N)$

Space

$O(N)$

C++ Execution & Sample Output

Core Function:

`isPossible()`

Validates whether a given max load `mid` can distribute all tasks across $\leq K$ servers using a single left-to-right pass.

📄 **Sample Input:** Tasks = {10, 20, 30, 40, 50}, K = 3

Optimal Max Load = 60

Server 1: 10 + 20 + 30 = 60

Server 2: 40

Server 3: 50





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Thank You